

# Limitations

## Selection / Self-Reporting Bias + LLM Standardization Impacts

Grad Cafe entries are voluntary and anonymous. The dataset is not a census of applicants, it's a self-selected sample of people who both knew that the site existed and decided posting was worth their time. It's likely that there is an over representation of a populus who are online more often, and possibly more anxious about the selection process. This data set has both a selection bias, which impacts the population that appears in the dataset at all, as well as a self-reporting bias which impacts those who did choose to post. We can expect that there is a whole population that lost interest in the process even if they were accepted, or were rejected and got demotivated to share negative data. In addition to there being a bias pool of submissions, not every field is filled in by every poster GPA, GRE, and nationality classification all have large fractions of nulls/blanks which impacts how stable our answers are when they're pulled from these small subsets of data. Self-reported means that even the field that are filled in are much more likely to contain typos, incorrectly remembered scores, inconsistent formatting of program/university names etc.

The Q8/Q9 discrepancy documented in *query\_results.pdf* is a different kind of limitation, it's a bias introduced by our own pipeline, specifically by the LLM standardization step that was meant to clean the data. In effect, the database doesn't just under-represent certain applicants for specific universities like Georgetown and Carnegie Mellon, our own cleaned fields can systematically undercount even the applicants who are present in the data, because the standardization step corrupted the very strings we search on. Georgetown University was rewritten as “George Town University” in every single instance we found, taking 318 matches in the original field down to zero in the LLM-generated field. Carnegie Mellon University was consistently misspelled as “Carnegie Melon University,” dropping from 750 matches to just 7, and this single error accounts for the entire gap between Q8's count of 30 and Q9's count of 26. Johns Hopkins University saw more modest, partial corruption, variants like “Johnson Hopkins University” and garbled strings were dropped from 753 to roughly 715 matches. Tellingly, Stanford, MIT, and the phrase “Computer Science” itself were essentially untouched by the same cleaning process, which shows this isn't a general degradation of data quality it's a specific, unpredictable failure tied to certain names

and strings. This matters because it means a query result being mathematically well-defined and reproducible, as ours are, is not the same as that result reflecting the true underlying counts. Taken together with the GRE contamination we found in Custom Q1, where 60.99% of the GRE Quantitative entries that even exist in the dataset turned out to be misplaced values or sentinel placeholders rather than real scores, the broader lesson is the same, a database can contain a well-formed number and still not tell you what you think it tells you. What our analysis can say with confidence is what these specific fields, as scraped, cleaned, and queried, actually contain. What it cannot say is that those numbers are a faithful measurement of the real population of Grad Café applicants, let alone graduate applicants as a whole.